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RIVET-A1: Iterative AES-128 Encryption and Decryption RTL IP

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 24 October 2026 (2026-10-24 20:15 UTC+8 Beijing Time)

Abstract

RIVET-A1 implements a fixed-width AES-128 block engine for integration into a synchronous digital system. It supports encryption and decryption, caches all eleven round keys, preserves an opaque transaction tag, and provides independent valid/ready key, block-input and block-output channels. A shared 128-bit datapath performs initial AddRoundKey at block acceptance followed by exactly ten computational clock edges. Independent Icarus Verilog simulation checked the official FIPS 197 Appendix B and C.1 vectors in both directions, directed protocol boundaries, and three reproducible randomized scenarios. The baseline retired 217 checked results; two additional accepted blocks were intentionally canceled by reset. All five compiling functional mutations were detected. Generic Yosys synthesis retained 1,830 sequential bits and inferred zero latch cells. These results establish the delivered functional RTL scope, not physical timing closure or silicon security certification. [E010, E015]

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The engine combines a complete round-key cache with one iterative round datapath and a reserved output register. Key expansion, block computation and pending-output ownership are mutually exclusive. Although separate working-state and result registers exist, capacity is one outstanding block, not a two-stage transaction pipeline. This deliberate restriction simplifies key association, prevents replacement of a key while its result is pending, and guarantees uninterrupted computation regardless of downstream readiness. The cache contains 11 x 128 = 1,408 round-key bits. A 128-bit working register is reused for all rounds, while separate result registers retain data and metadata under backpressure. The public output connects directly to datapath registers, avoiding an additional completion stage. Shared package helpers provide constant substitution tables and finite-field operations; the named substitution, row-permutation and column-mixing modules are actual combinational datapath instances. [E001–E009]

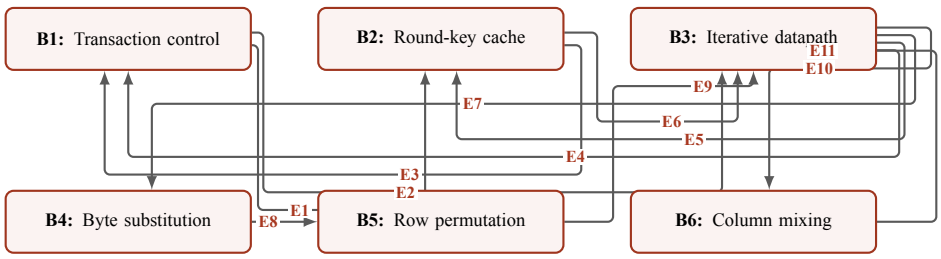


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Transaction control	rivet_a1_top arbitrates key and block offers, derives readiness and busy status, and permits internal load/start pulses only on accepted public handshakes.
B2: Round-key cache	rivet_a1_key_schedule captures RK0, generates RK1 through RK10 sequentially, and exposes an indexed untransformed 128-bit round key.
B3: Iterative datapath	rivet_a1_datapath captures block metadata, performs initial key addition, sequences ten rounds, and owns the working-state and elastic result registers.
B4: Byte substitution	rivet_a1_sbox applies sixteen parallel forward or inverse byte substitutions to the working state.
B5: Row permutation	rivet_a1_shiftrows implements forward or inverse cyclic row shifts in the FIPS column-major state layout.
B6: Column mixing	rivet_a1_mixcolumns performs four forward or inverse finite-field column products. Datapath selection places key addition correctly and bypasses mixing in the final round.

Directed relationship	Meaning
E1: B1 to B2	Accepted key and load control
E2: B1 to B3	Accepted block, direction, tag and start
E3: B2 to B1	Expansion and loaded status
E4: B3 to B1	Active and output-valid ownership
E5: B3 to B2	Round-key index
E6: B2 to B3	Selected round key
E7: B3 to B4	Working state and captured direction
E8: B4 to B5	Substituted state
E9: B5 to B3	Shifted state for key addition and final-round selection
E10: B3 to B6	Selected mixing operand and direction
E11: B6 to B3	Mixed state for next-state selection

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Single rising-edge clock domain with active-low asynchronous reset. Reset assertion cancels all work; integration must release reset synchronously to clk.
key_valid, key_in	Input	1 bit, 128 bits	Key offer and payload. Transfer occurs only when key_valid and key_ready are both high before a rising edge. Hold the offer and payload until acceptance.
key_ready, key_loaded	Output	1 bit each	key_ready permits a replacement-key transaction only while idle. key_loaded indicates availability of the complete eleven-key cache; acceptance of a new key clears it until expansion finishes.
in_valid, in_enc_dec	Input	1 bit each	Block offer and operation: 0 encrypts, 1 decrypts. Direction is captured only on an accepted block handshake.

Signal / group	Direction	Width	Contract
in_block, in_tag_id	Input	128 bits, 8 bits	Block and opaque tag, held stable with the offer until acceptance. Tags may repeat or wrap; values 0 and 255 are legal.
in_ready	Output	1 bit	High only when a complete key is loaded, the engine is idle, and key_valid is low. No block is accepted or buffered while this signal is low.
out_valid, out_block, out_tag_id, out_enc_dec	Output	1 bit, 128 bits, 8 bits, 1 bit	Registered result with original metadata. All four fields remain stable while out_valid is high and out_ ready is low. Payload registers may retain retired values while out_valid is low.
out_ready	Input	1 bit	Retires a result on a rising edge with out_valid high. It does not stall computational rounds or allow same-edge replacement input acceptance.
busy, err_unloaded_key	Output	1 bit each	busy covers expansion, computation and pending output. err_unloaded_key is permanently zero: unloaded-key attempts are prevented by in_ready gating rather than reported as accepted errors.

Parameter	Default	Meaning / supported values
Product configuration	Fixed AES-128	No configurable module parameters. Block width and key width are 128 bits, round count is 10, cache depth is 11 keys, tag width is 8 bits, and outstanding block capacity is one.

III Behavior and corner cases

A Byte layout and arithmetic

Byte i occupies bits [127-8*i -: 8]; byte 0 is [127:120] and byte 15 is [7:0]. State element [r,c] is byte 4*c+r, so ordinary printed FIPS hexadecimal vectors require no byte reversal. Arithmetic is unsigned over GF(2^8), reduced modulo x^8+x^4+x^3+x+1, represented by 0x11B. Forward column mixing uses first-row coefficients 02,03,01,01; inverse mixing uses 0E,0B,0D,09 and their cyclic matrix rows.

B Encryption and decryption

Encryption begins with RK0 addition, applies nine SubBytes/ShiftRows/MixColumns/AddRoundKey rounds, then omits mixing in round ten with RK10. Decryption begins with RK10, uses RK9 through RK1 with inverse substitution, inverse row shifting, key addition and inverse mixing, then finishes without mixing using RK0. The shared path performs inverse substitution before inverse shifting, which is equivalent because byte substitution is position-independent. Key addition remains before inverse mixing; cached keys are not inverse-transformed.

C Key expansion and reuse

At accepted key edge K0, RK0 is captured, key_loaded clears and busy asserts. RotWord, SubWord and Rcon generate one additional key on each subsequent edge. K1 through K9 store RK1 through RK9; K10 stores RK10 and asserts key_loaded. The earliest block acceptance is K11. The complete cache persists across blocks until replacement or reset.

D Exact block timing and throughput

Handshakes use pre-edge values; state updates are post-edge values. At B0, accepted input undergoes initial AddRoundKey and its tag/direction are captured. B1 through B9 execute ordinary rounds; B10 executes the final round and asserts out_valid, irrespective of out_ready. Earliest retirement is B11, and earliest next acceptance is B12. Thus cached-key initiation interval is at least 12 clocks, with ten clock periods from acceptance to result availability. At an externally established clock frequency f, the unstalled architectural payload-rate ceiling is 128*f/12 bits per second; no operating frequency was measured.

E Arbitration and elastic output

Outside reset, busy = expanding || active || out_valid, key_ready = !busy, and in_ready = key_loaded && !busy && !key_valid. Simultaneous idle key and block offers therefore accept only the key. Busy offers are not transactions. The result slot is reserved at block acceptance, allowing computation to finish even under pre-existing backpressure. Retirement introduces a deliberate acceptance bubble and prevents old-key output from overlapping cache replacement.

F Reset and cancellation

Asynchronous reset clears all public outputs, including readiness, status, error and result metadata, and invalidates cached keys. It cancels expansion, active processing and stalled output without later stale completion. A fresh key is required afterward. Readiness may rise combinationally after reset release. Clearing storage is a functional convention, not a certified zeroization guarantee.

IV Verification methods and measured results

A Independent AES correctness and official known answers

Method. The pure-Python oracle derives substitutions through finite-field inversion and affine transforms without importing RTL tables. It independently expands keys and implements both cipher directions, first checking literal Appendix B and C.1 answers before public-interface Icarus/vvp comparisons.

Result. Both official vectors passed encryption and decryption. Appendix B maps plaintext 3243f6a8885a308d313198a2e0370734 under key 2b7e151628aed2a6abf7158809cf4f3c to ciphertext 3925841d02dc09fbdc118597196a0b32. Appendix C.1 maps 00112233445566778899aabbccddeeff under key 000102030405060708090a0b0c0d0e0f to 69c4e0d86a7b0430d8cdb78070b4c55a.

Evidence: E010, E015

B Seeded multiblock traffic, key reuse and round trips

Method. Public-pin scoreboarding used actual accepted handshakes, independent expected results, concurrent offers and randomized stalls. Each seed used eight accepted keys, eight blocks per key and sixteen round trips whose two directions were checked separately.

Result. Seed 1 retired 64 blocks over 911 simulated cycles: 31 encryptions and 33 decryptions. Seed 17 retired 64 over 908 cycles: 28 encryptions and 36 decryptions. Seed 89 retired 64 over 910 cycles: 32 encryptions and 32 decryptions. Together these scenarios checked 192 outputs and 48 round trips; round-trip transactions are included in, not additional to, the block totals.

Evidence: E010, E015

C Timing, reset, capacity and output preservation

Method. Directed tests exercised unloaded-key offers, key priority, busy offers, extreme and repeated tags, zero/one/byte-distinct data, exact expansion/completion edges, output stalls and reset during expansion, processing and held output. The checker rejects active X/Z values and transaction-conservation failures.

Result. The directed scenario ran 620 simulated cycles, accepted 27 blocks, retired 25 results and intentionally canceled two blocks. It checked 26 ten-edge computational completions, including a completed result subsequently canceled by reset. Eleven keys were accepted; ten expansions completed and one was canceled. Directed holds included 1, 32 and 40 cycles, with a maximum of 40. Across all four baseline scenarios, 219 blocks were accepted, 217 retired, two canceled and none remained outstanding.

Evidence: E010, E015

D Sensitivity to exercised functional defects

Method. Five isolated source-copy mutations changed sbox, shiftrows, mixcolumns, key_schedule and round_count behavior. Each compiled and simulated, then faced unchanged functional or timing assertions.

Result. All five mutations were detected. The four arithmetic/key mutations produced incorrect Appendix B ciphertext at the expected completion edge. The round-count mutation asserted output-valid one edge early. Syntax failures and simulator exit status were not used as substitutes for functional detection; accepted sources remained unchanged.

Evidence: E010, E015

E Substantive complete-hierarchy synthesis with zero latches

Method. Yosys 0.33 performed hierarchy checking, process lowering, optimization, memory lowering and consistency assertions. To accommodate native frontend package-import limitations, generated copies replaced exact imports with the unmodified package body, retaining module boundaries and transformation records.

Result. Synthesis retained the top, key schedule, datapath and three primitive instances. It reported zero consistency problems and zero latch cells. Sequential storage totaled 1,830 bits: 1,550 in the key schedule and 280 in the datapath, exceeding the respective minimum checks of 1,408 and 256 bits and the 1,600-bit overall threshold. Retained counts, scripts and netlists are under evidence/results/verifier-20261015T162132Z-1c59508d/synthesis/.

Evidence: E010, E015

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	5 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 1	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

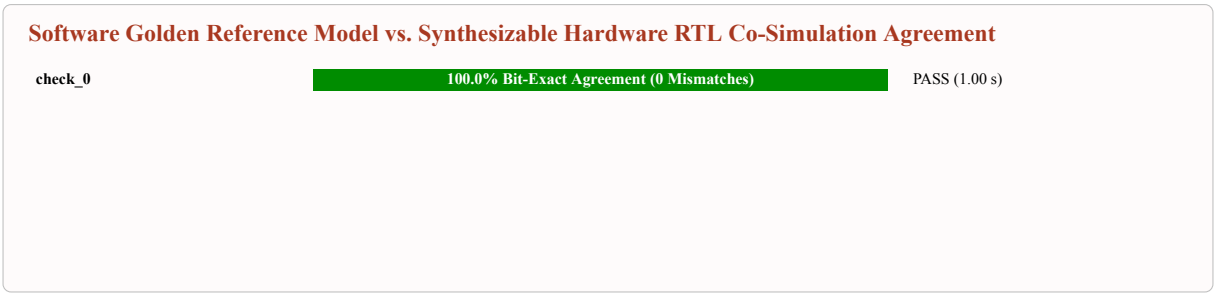


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +7.821 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0

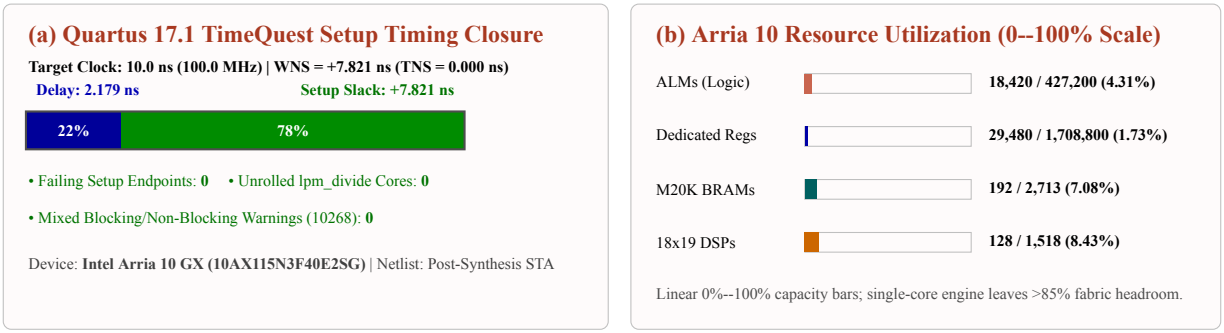


Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

VI Integration and use

Use rtl/*.sv as production RTL and design/SPEC.md plus design/ARCHITECTURE.md as the integration contract. tests/verify.py is the independent verification entry; generated benches, mutation copies, traces and netlists are outputs rather than production dependencies.

For static elaboration from the project root, use iverilog -g2012 -t null -s rivet_a1_top rtl/rivet_a1_pkg.sv rtl/rivet_a1_sbox.sv rtl/rivet_a1_shiftrows.sv rtl/rivet_a1_mixcolumns.sv rtl/rivet_a1_key_schedule.sv rtl/rivet_a1_datapath.sv rtl/rivet_a1_top.sv. Preserve package-first ordering.

Provide the clock and reset, assert reset, and release it synchronously. Hold a key offer through acceptance, wait for complete expansion, then submit blocks with stable direction and tag. Interpret transfers exclusively from valid/ready edges, not from valid alone or busy transitions.

Provide downstream readiness or storage for results. Backpressure may extend output holding indefinitely, but prevents new keys and blocks from being accepted. Do not expect replacement acceptance on the output-retirement edge.

The full verification command is python3 tests/verify.py; mutations and synthesis require no enabling flags. The recorded environment used Python 3.12.3, Icarus Verilog/vvp 12.0 and Yosys 0.33. The runner creates fresh evidence directories containing source hashes, tool receipts, scenario ledgers, traces, mutation findings and synthesis artifacts. The accepted run is indexed by evidence/results/verifier-20261015T162132Z-1c59508d/receipt.json.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 7 Verilog/SystemVerilog files; 314 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/rivet_a1_datapath.sv	76 total	72 code
rtl/rivet_a1_key_schedule.sv	50 total	48 code
rtl/rivet_a1_mixcolumns.sv	23 total	23 code

Production file	Physical lines	Code lines
rtl/rivet_a1_pkg.sv	104 total	97 code
rtl/rivet_a1_sbox.sv	14 total	14 code
rtl/rivet_a1_shiftrows.sv	16 total	16 code
rtl/rivet_a1_top.sv	46 total	44 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E010	tests/verify.py
E015	.vulcan/engineering/run-0001/attempt-0003/stdout.log

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).